

STAGE 9: DETERMINISTIC DOCX GENERATION ENGINE VERIFICATION REPORT

System Version: 1.0.0 | Verification Date: September 29, 2026 | Author: ARM Quality Assurance & Core Document Engine Team | Overall Status: **PASS**
(READY FOR STAGE 10)

1. EXECUTIVE SUMMARY & STAGE OBJECTIVE

Stage 9 introduces the deterministic document generation engine for ARM (AI Academic Report Assistant). The objective of Stage 9 is to synthesize: **Locked College Template (Stage 4) + Approved Report Plan (Stage 6) + Generated Report Content (Stage 7) + Stage 8 Verification Results (Quality Gates) + Verified Evidence & Citations** into a valid, OpenXML-compliant DOCX report document without allowing AI to dictate document styling, geometry, margins, or fonts. Final formatting is deterministic and strictly governed by the locked TemplateSchema.

2. KEY ARCHITECTURAL GUARANTEES & VERIFIED RESULTS

Architectural Pillar	Engine Implementation & Quality Gate	Verification Status
Locked Template Authority	Rejects compilation if template is not locked (status='locked'). TemplateSchema v1.0.0 enforces page size, margins, font family, line spacing, heading levels, and headers/footers.	VERIFIED & LOCKED
Original Template Immutability	Original uploaded template file is NEVER mutated. Pre-generation SHA-256 is recorded and compared against post-generation SHA-256. Automated test verifies exact hash preservation.	VERIFIED (SHA-256 Intact)
Approved ReportPlan Gate	Requires approved ReportPlan (is_approved=True). Sections are mapped deterministically according to the authoritative locked template sequence.	VERIFIED & ENFORCED
Stage 8 Quality Gate Enforcement	Stage 8 verification results are checked before document compilation. Blocking claims (e.g. invalid citation, fabricated figures) reject final compilation with HTTP 422, or generate as REVIEW_REQUIRED with force=True.	VERIFIED & ENFORCED
Structural DOCX Validation & Reopen Test	DocumentValidator verifies ZIP container integrity, essential OpenXML parts ([Content_Types].xml, _rels/.rels, word/document.xml), and reopens the generated DOCX via python-docx to verify sections, paragraphs, and tables.	VERIFIED (python-docx Reopened)
Multi-Tenant IDOR Security	Project ownership is verified prior to document compilation and download. Unauthorized cross-tenant compile or download requests from User B on User A's report return HTTP 403 Forbidden.	VERIFIED (HTTP 403 Enforced)
Private Storage & File Sanitization	Documents are persisted to private storage at users/{uid}/projects/{pid}/reports/{rid}/versions/{vid}/report.docx. Filenames are sanitized against path traversal.	VERIFIED & SECURED

3. AUTOMATED TEST SUITE EXECUTION RESULTS

Test Suite / Stage	Scope & Subsystems Covered	Tests	Result
Stage 1 — Database Foundation	PostgreSQL schema, RLS policies, profiles, generation_jobs	7 / 7	100% PASS
Stage 2 — Template Ingestion	DOCX/PDF upload, MIME validation, private storage	10 / 10	100% PASS
Stage 3 — Template Intelligence	Deterministic OpenXML AST parsing, TemplateSchema normalization	9 / 9	100% PASS
Stage 4 — Template Review + Lock	Patching template schema, hierarchy validation, lock immutability	10 / 10	100% PASS
Stage 5 — Project Evidence Locker	Fact storage, project members, file evidence, readiness indicator	8 / 8	100% PASS
Stage 6 — AI Report Planner	ReportPlan generation, section breakdown, Gemini integration	7 / 7	100% PASS
Stage 7 — Report Content Generator	Section-by-section Gemini generation, grounding status, versioning	10 / 10	100% PASS
Stage 8 — Citation Verifier	Claim extraction, evidence mapping, anti-fabrication, quality gates	12 / 12	100% PASS
Baseline Services & Endpoints	Health check, document assembler baseline, template parser	12 / 12	100% PASS
Stage 9 — Deterministic DOCX Engine	DocumentAssembler, DocumentValidator, SHA-256 immutability, IDOR, reopen test	11 / 11	100% PASS
TOTAL INTEGRATED TEST SUITE	Zero regressions across all stages 1 through 9	96 / 96	100% PASSED

4. STAGE 9 DEDICATED VERIFICATION SUITE BREAKDOWN

Test Identifier	Tested Requirement	Verified Behavior	Result
test_mandatory_document_assembly_and_reopen_test	Mandatory Document Test (Sec 55)	Assembled full document with headings, subheadings, paragraphs, bullet lists, numbered lists, styled table with headers & caption, blockquote, and references. Successfully reopened via python-docx.	PASS
test_original_template_immutability_sha256	Template Immutability (Sec 4, 40)	Master template file SHA-256 checksum recorded before and after generation. Proved original template file remains strictly untouched.	PASS
test_locked_template_enforcement	Locked Template Gate (Sec 5)	Compilation request with unlocked/draft template strictly rejected with HTTP 400 Bad Request.	PASS
test_approved_report_plan_enforcement	Approved Plan Gate (Sec 6)	Compilation request with missing or unapproved report plan rejected with HTTP 400 Bad Request.	PASS
test_stage8_verification_quality_gate_blocks_generation	Quality Gate Gate (Sec 7, 8)	Blocking issues from Stage 8 reject generation with HTTP 422 unless force=True is provided, which compiles document in REVIEW_REQUIRED draft status.	PASS
test_different_templates_distinct_formatting	Template Fidelity (Sec 57)	Template A (Times New Roman, 12pt, 31.75mm margins) vs Template B (Arial, 10.5pt, 20mm margins) generated distinct typography and page geometries.	PASS
test_different_projects_distinct_content	Project Isolation (Sec 59)	Attendance project DOCX contains facial recognition content; Agriculture project DOCX contains irrigation content. Zero cross-project data bleed.	PASS
test_multi_tenant_idor_protection	IDOR Protection (Sec 45, 60)	User B denied compile and download access to User A's project document with HTTP 403 Forbidden.	PASS
test_path_traversal_and_filename_sanitization	Security & Filenames (Sec 33, 48)	Malicious project titles with directory traversal characters safely sanitized.	PASS
test_generation_job_lifecycle	Generation Jobs (Sec 36)	Job lifecycle progresses through validating_inputs -> assembling_document -> validating_docx -> completed (100%).	PASS
test_api_compile_and_download_flow	Full API Flow (Sec 51)	POST /compile-docx compiles DOCX; GET /download-docx streams DOCX binary with valid Content-Disposition headers.	PASS

5. FRONTEND VERIFICATION & PRODUCTION BUILD

- Frontend Status:
- **TypeScript Verification:** tsc -p apps/web/tsconfig.json --noEmit completed with **0 errors**.
 - **ESLint Verification:** npm run lint completed with **0 errors and 0 warnings**.
 - **Next.js Production Build:** npm run build successfully compiled all **26/26 routes** with static optimization.
 - **UI Component Integration:** Created DocumentCompiler (apps/web/src/components/arm/document-compiler.tsx) integrated into /reports with live job progress stepper, quality gate badges, validation details, and secure DOCX download.

6. STAGE BOUNDARY & SIGN-OFF

STAGE 9 COMPLETION DECLARATION

Deterministic DOCX Generation Engine is fully implemented, verified, and locked.

- **Stage 9 Status:** **PASS**
- **Total Tests Passed:** 96 / 96 (0 Regressions across Stages 1–9)
- **Original Template Immutability:** VERIFIED (SHA-256 Unchanged)
- **DOCX OpenXML Integrity & Reopen Test:** VERIFIED (python-docx)
- **Stage Boundary Enforcement:** PDF conversion, PDF preview, and visual comparison have NOT been implemented (strictly deferred to Stage 10).
- **READY FOR STAGE 10 (PDF GENERATION):** **YES**